

IPv4 Multicast

Tecnologie e Servizi di Rete

Computer Network Technologies and Services

Internet Architecture and Technologies

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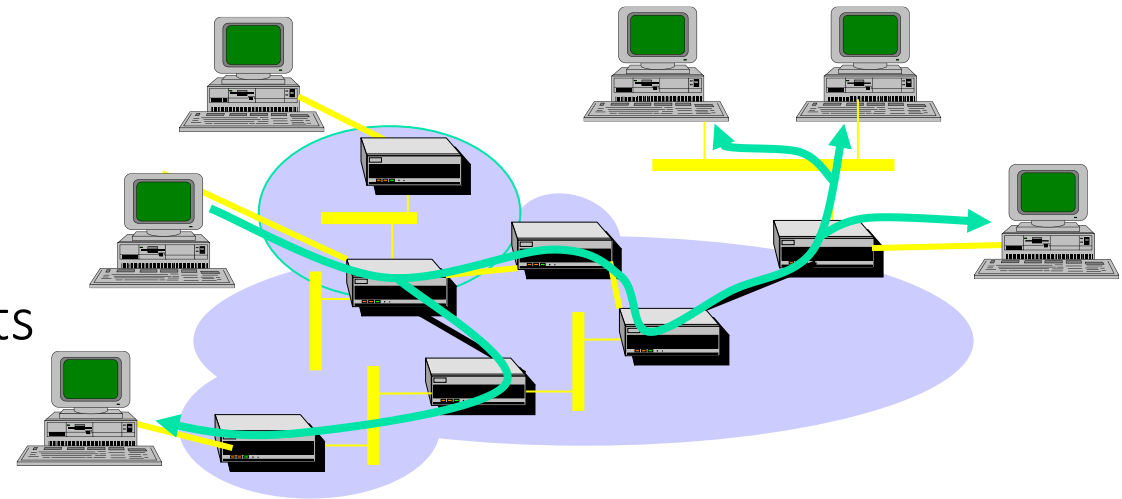
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Underlying Concept

- Packets are routed from a source to **multiple** destinations, collectively identified by an IP **multicast address**
 - Key for group communication
 - e.g., videoconferencing, video broadcasting
- An IP Multicast Address identifies a group of hosts
 - Hosts **actively** join/leave a group
 - Hosts can selectively choose which source(s) in the group they receive traffic from
 - Routers **keep track** of which groups have active listeners and route packets to individual members



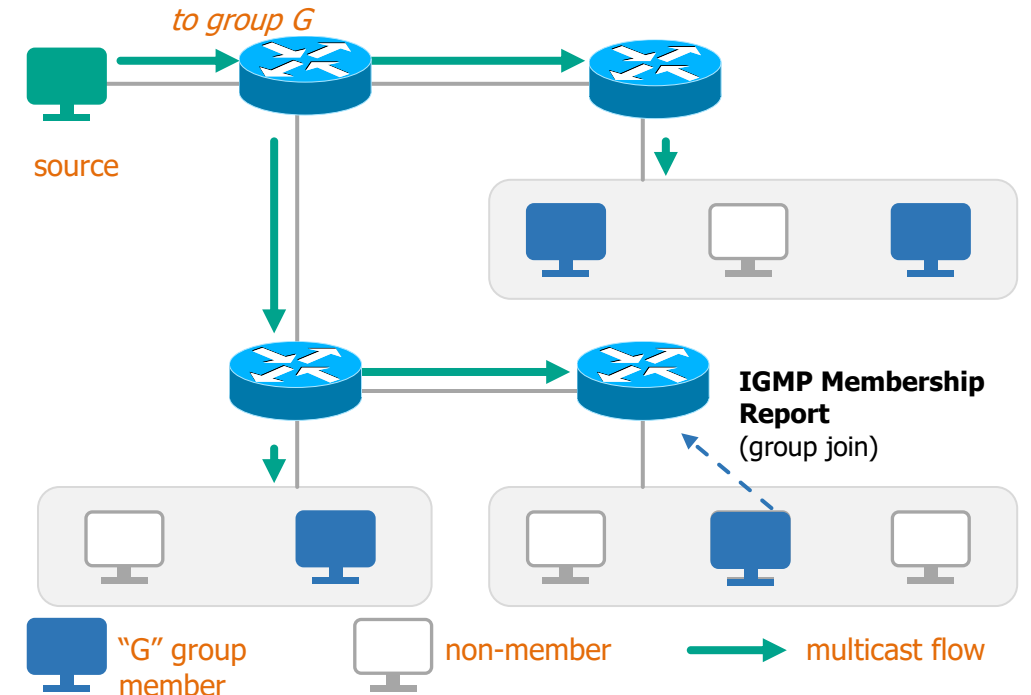
Multicast addressing

- Class D addresses
 - Begin with 1110
 - 224.0.0.0 - 239.255.255.255
- Address identifies a *host group*
 - Packet is delivered to all hosts in the group
 - Anywhere in the network (across routers), where multicast routing is enabled
- Notable ranges (IANA, RFC 5771):
 - 224.0.0.0/24: link-local control, never forwarded (224.0.0.1 all hosts, 224.0.0.2 all routers, 224.0.0.22 IGMPv3, 224.0.0.251 mDNS)
 - 232.0.0.0/8: source-specific multicast (RFC 4607)
 - 239.0.0.0/8: administratively scoped, the “private” multicast range (RFC 2365)



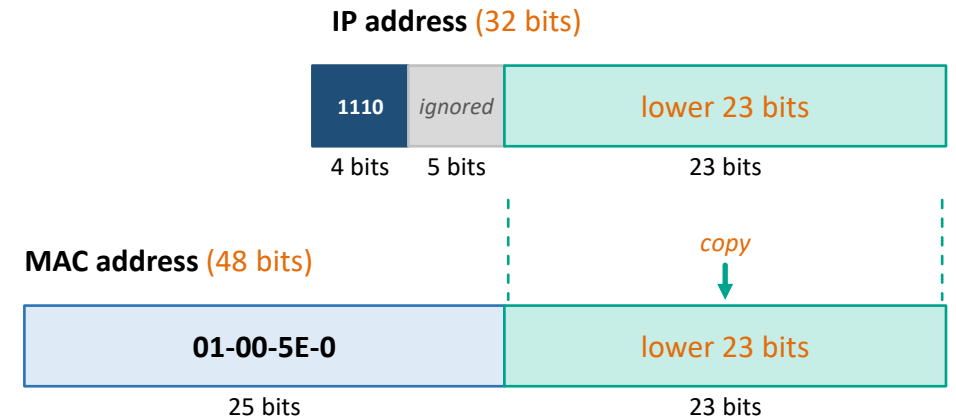
Host Group Membership

- Hosts join and leave dynamically
- **IGMP** (Internet Group Management Protocol) protocol (encapsulated in IPv4 datagrams) handles membership within a local network
 - Used by hosts and routers; IP datagrams with TTL = 1
- Recipients of a multicast datagram establish which hosts receive a packet
 - In unicast, it is the source
 - Controlling traffic reach is more difficult



Within a Local IEEE 802 Network

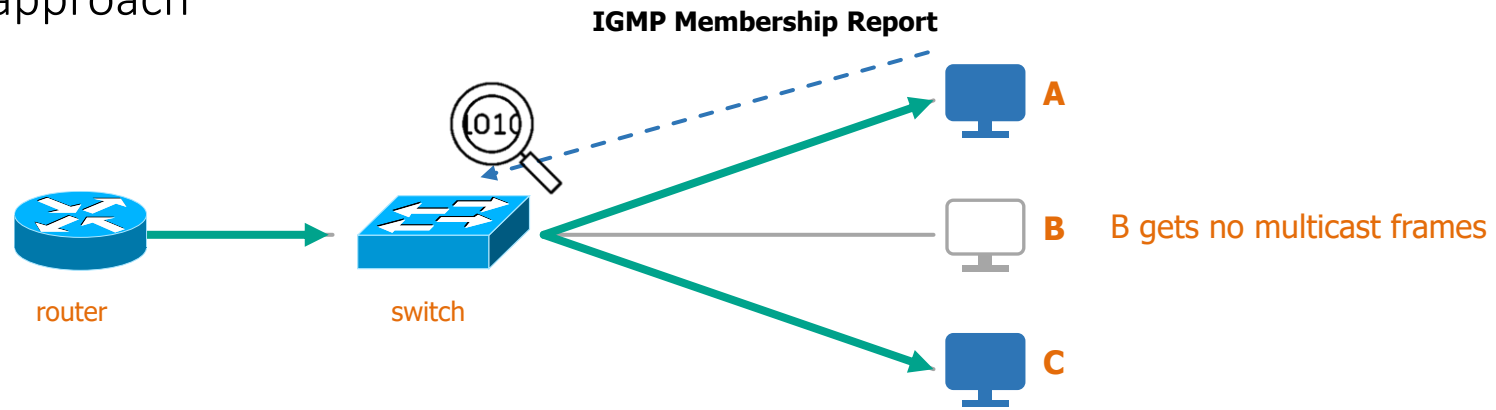
- Group delivery delegated to lower level (MAC)
- IP multicast address mapped to a MAC multicast address (RFC 7042), built as:
 - 25 fixed most significant bits: 01-00-5E-0
 - 23 least significant bits are taken from the corresponding bits in the IP address



- MAC address collisions are resolved at the IP layer

Within a Local IEEE 802 Network

- Interface card configured to receive MAC multicast
- Recipient-initiated group join through IGMP
- Switch support through *IGMP Snooping*
 - Multicast delivery only on ports connected to a group member
 - requires cross-layer approach



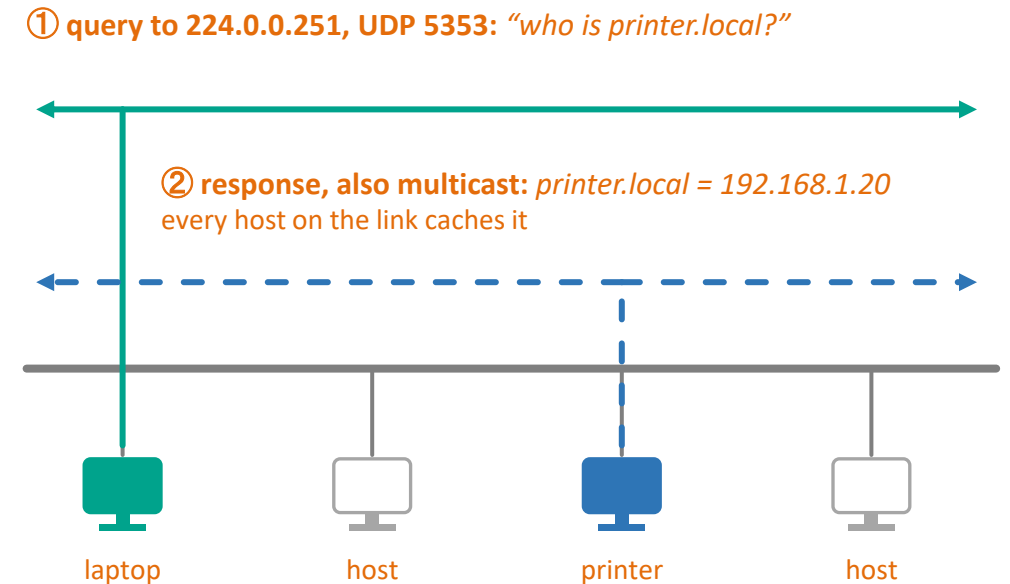
Beyond single networks

- Routers manage host groups on each LAN using **IGMP**
 - hosts send **IGMP Membership Report messages** to 224.0.0.22 (multicast address for routers)
 - Routers periodically send **IGMP General Query** messages to discover which groups are active on the Local Network
- Routers announce host groups to others
 - Multicast routing protocols
- Routers build a distribution tree for each host group
 - To all LANs with at least a member



Example: Multicast DNS (mDNS)

- Name resolution on a single link, without a DNS server (RFC 6762)
 - Standard DNS messages sent to 224.0.0.251
 - Names end in .local
- Responses are usually multicast too: every host on the link can cache them
- Before using a name, a host probes it then announces it
- Paired with DNS-SD (RFC 6763) it also discovers services: printers, AirPlay, Chromecast
 - Apple Bonjour, Avahi on Linux, native since Windows 10
 - E.g.: “any Airplay receivers out there?”
- 224.0.0.0/24 is link-local: routers never forward it, so discovery stops at the subnet



On large Wi-Fi networks every announcement reaches every device: campuses often filter or proxy mDNS

Deployment status

- Not widely supported
- Not suitable for common traffic control/engineering practice in the Internet at large
 - Inter-domain multicast traffic is practically absent
- Mostly limited to controlled environments
 - Video broadcasting within an ISP network
 - Local network service discovery
 - Data center and enterprise networks

